

E - Tree Growing

分值: 900 分

Problem Statement

You are given a tree with N vertices numbered 1 to N . Here, N is at least 2. The i -th edge ($1 \leq i \leq N - 1$) connects vertices u_i and v_i .

给定一棵有 N 个顶点的树，顶点编号为 1 到 N 。保证 $N \geq 2$ 。第 i 条边 ($1 \leq i \leq N - 1$) 连接顶点 u_i 和 v_i 。

Perform the following operation on this tree exactly K times. The k -th operation ($1 \leq k \leq K$) is as follows.

对这棵树恰好执行 K 次操作。第 k 次操作 ($1 \leq k \leq K$) 的内容如下。

- Choose one of the $N - 2 + k$ edges of the tree at that point uniformly at random. Let u and v be the vertices connected by the chosen edge. Prepare a new vertex $N + k$, add edges connecting vertices u and $N + k$ and connecting vertices v and $N + k$, then delete the chosen edge.
- 从当前树的 $N - 2 + k$ 条边中等概率随机选择一条。设所选边连接的顶点为 u 和 v 。新建一个顶点 $N + k$ ，添加边 $u-N + k$ 和 $v-N + k$ ，然后删除选中的边。

Note that for $k = 1, 2, \dots, K$, the tree will have $N + k$ vertices after the k -th operation.

注意，对于 $k = 1, 2, \dots, K$ ，第 k 次操作结束后，树将有 $N + k$ 个顶点。

Find the expected value, modulo 998244353, of $\sum_{1 \leq i < j \leq N+K} \text{dist}(i, j)$ after K operations. Here, $\text{dist}(i, j)$ denotes the distance between vertices i and j in the tree.

求 K 次操作后 $\sum_{1 \leq i < j \leq N+K} \text{dist}(i, j)$ 的期望值，结果对 998244353 取模。其中 $\text{dist}(i, j)$ 表示树中顶点 i 和 j 之间的距离。

You are given T test cases; solve each of them.

共有 T 组测试数据，你需要处理每组数据。

Definition of expected value 998244353

期望值对 998244353 取模的定义

It can be proved that the expected value to be found is always a rational number. Moreover, under the constraints of this problem, when this rational number is expressed as an irreducible fraction $\frac{P}{Q}$, it can be proved that $Q \not\equiv 0 \pmod{998244353}$. Thus, there is a unique integer R satisfying $R \times Q \equiv P \pmod{998244353}, 0 \leq R < 998244353$. Find this R .

可以证明所求的期望值总是有理数。此外，在本题的约束条件下，将该有理数化为既约分数 $\frac{P}{Q}$ 时，可证 $Q \not\equiv 0 \pmod{998244353}$ 。因此存在唯一的整数 R 满足 $R \times Q \equiv P \pmod{998244353}, 0 \leq R < 998244353$ ，你需要求出这个 R 。

Constraints

- $1 \leq T$
- $2 \leq N$
- The sum of N over all test cases is at most 3×10^5 .
- $1 \leq K$
- The sum of K over all test cases is at most 10^6 .
- $1 \leq u_i < v_i \leq N$
- The given graph is a tree.
- All input values are integers.
- $1 \leq T$
- $2 \leq N$
- 所有测试数据的 N 之和不超过 3×10^5 。
- $1 \leq K$
- 所有测试数据的 K 之和不超过 10^6 。
- $1 \leq u_i < v_i \leq N$
- 给定的图是一棵树。
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```
T
case1
case2
⋮
caseT
```

Each test case is given in the following format:

每组测试数据按照以下格式给出：

```
N K
u1 v1
u2 v2
⋮
uN-1 vN-1
```

Output

Output the answers for the test cases in order, separated by newlines.

按顺序输出每组测试数据的答案，相邻答案用换行分隔。

Sample Input 1

```
3
4 2
1 2
1 3
1 4
4 5
1 2
2 3
3 4
6 13
```

```
1 2
1 3
2 4
2 5
3 6
```

Sample Output 1

```
499122208
120
713032723
```

Consider the first test case.

考虑第一组测试数据。

For example, the two operations proceed as follows.

例如，两次操作的过程如下。

- Operation 1: The edge connecting vertices 1 and 2 is chosen. After adding edges connecting vertices 1 and 5 and connecting vertices 2 and 5, the edge connecting vertices 1 and 2 is deleted.
- Operation 2: The edge connecting vertices 2 and 5 is chosen. After adding edges connecting vertices 2 and 6 and connecting vertices 5 and 6, the edge connecting vertices 2 and 5 is deleted.
- 第 1 次操作：选中连接顶点 1 和 2 的边。添加边 1-5 和 2-5 后，删除连接顶点 1 和 2 的边。
- 第 2 次操作：选中连接顶点 2 和 5 的边。添加边 2-6 和 5-6 后，删除连接顶点 2 和 5 的边。

In this case, the value of $\sum_{1 \leq i < j \leq N+K} \text{dist}(i, j)$ is 32.

这种情况下， $\sum_{1 \leq i < j \leq N+K} \text{dist}(i, j)$ 的值为 32。

The value of $\sum_{1 \leq i < j \leq N+K} \text{dist}(i, j)$ is 31 with probability $\frac{1}{2}$ and 32 with probability $\frac{1}{2}$.

$\sum_{1 \leq i < j \leq N+K}$ dist(i, j) 的值为 31 的概率是 $\frac{1}{2}$, 为 32 的概率是 $\frac{1}{2}$ 。

